

Customer Sales Analysis Report

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Project: Week 5 – Customer Sales Analysis

Executive Summary

This report presents a complete customer sales analysis using Python and Pandas. The project includes data cleaning, transformation, aggregation, filtering, pivot tables and visualizations to identify sales trends, top customers, regional performance and product performance.

Project Overview

Objective:

- Analyze customer purchasing patterns.
- Identify top customers.
- Analyze monthly, regional and product-wise sales.
- Generate business insights and recommendations.

Tools & Technologies

Python

Pandas

Matplotlib

Jupyter Notebook

CSV Dataset

Dataset Description

Dataset columns: Date, Product, Quantity, Price, Customer_ID, Region, Total_Sales.

The Date column was converted to datetime and Year, Month and Day were extracted.

Data Cleaning & Preparation

Loaded the CSV file, checked data types, verified missing values, converted Date to datetime, created Year/Month/Day columns and validated numeric columns.

Exploratory Data Analysis (EDA)

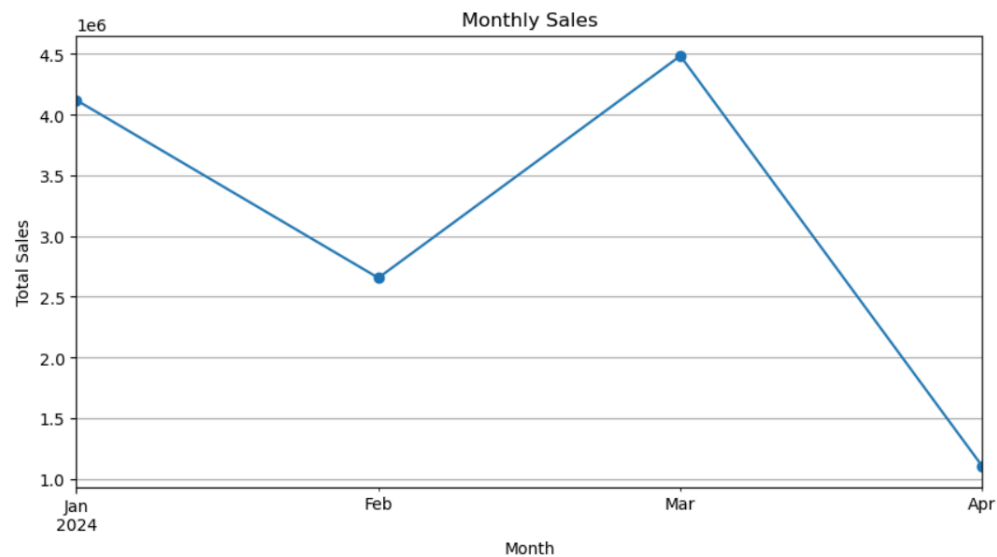
Performed total revenue analysis, monthly sales trend, region-wise sales, product-wise sales, top customer analysis and revenue distribution.

Data Manipulation Using Pandas

Applied groupby(), sum(), mean(), count(), sorting, filtering (AND/OR), datetime extraction and pivot tables.

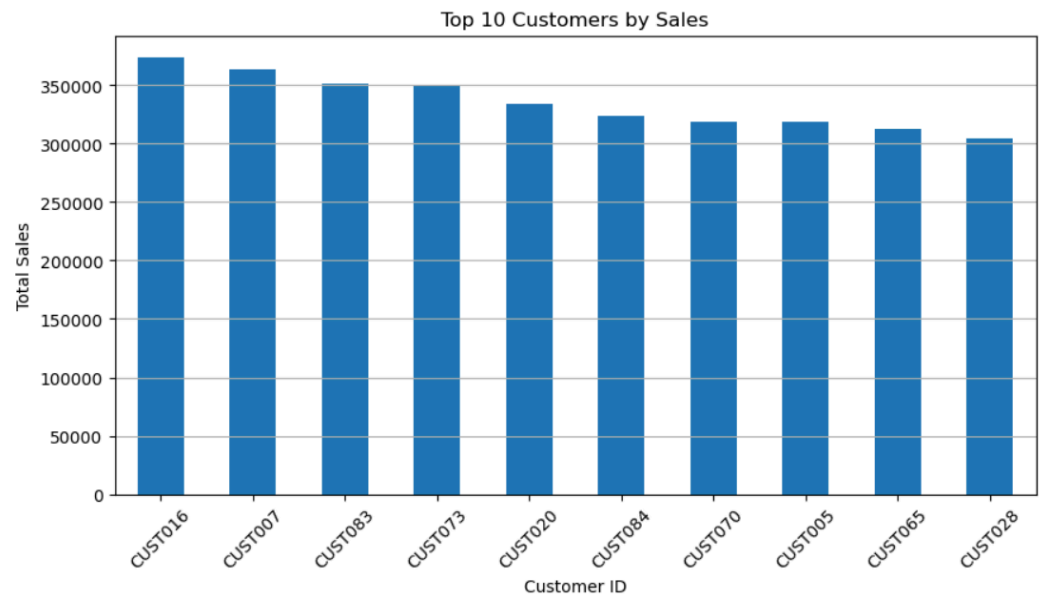
Visualizations

Monthly Sales Trend



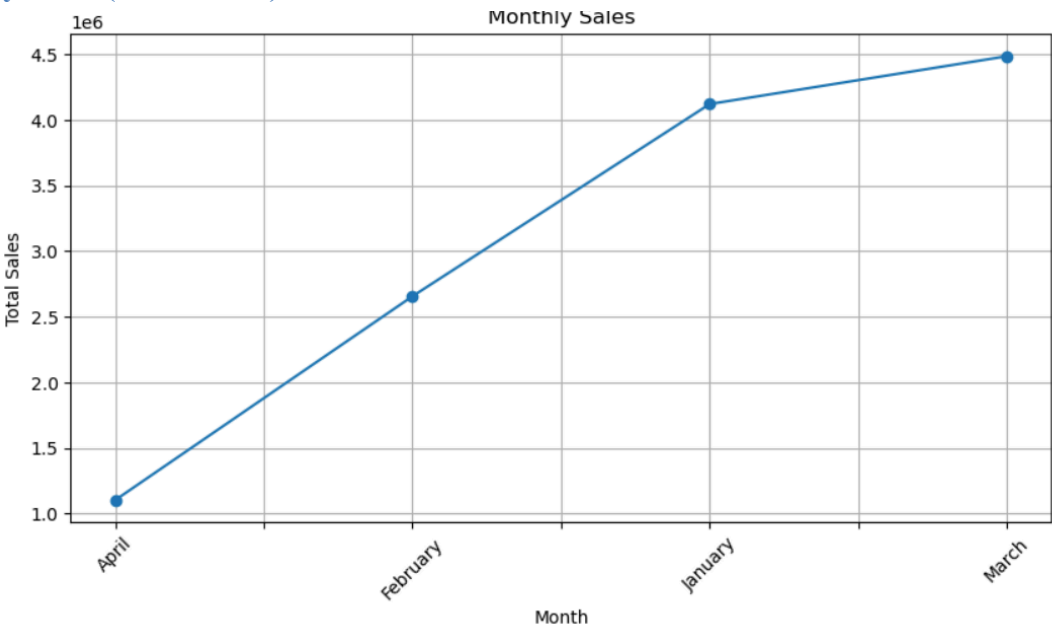
Analysis: Sales peak in March and decline sharply in April.

Top 10 Customers



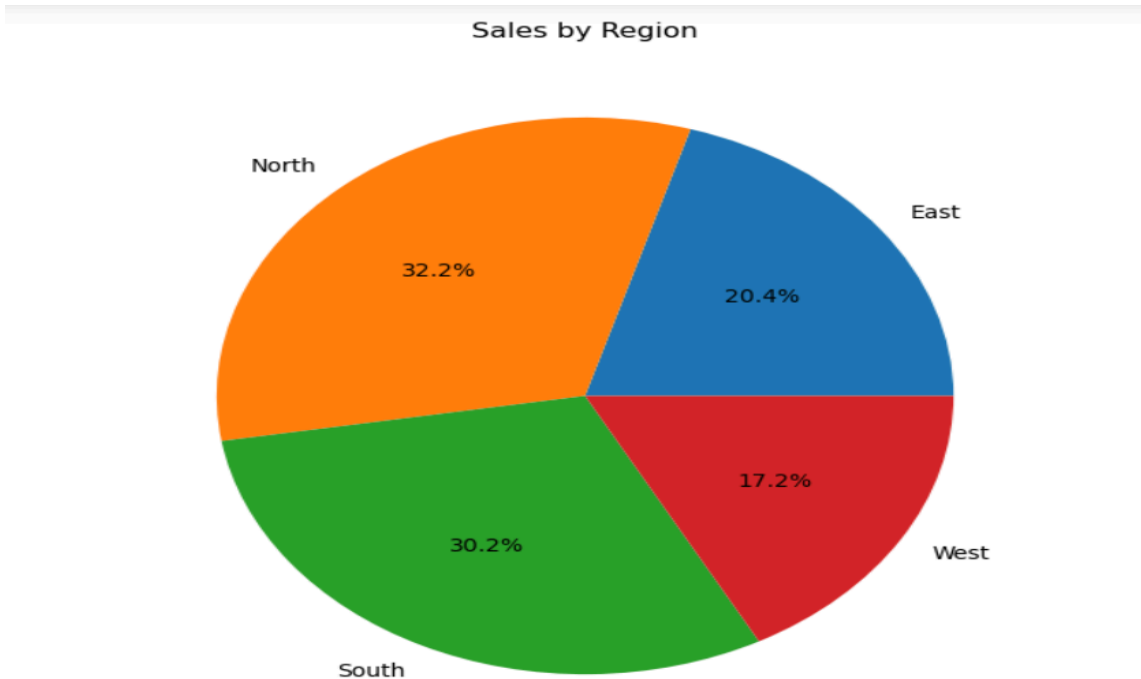
Analysis: Shows the highest revenue-generating customers.

Monthly Sales (Alternative)



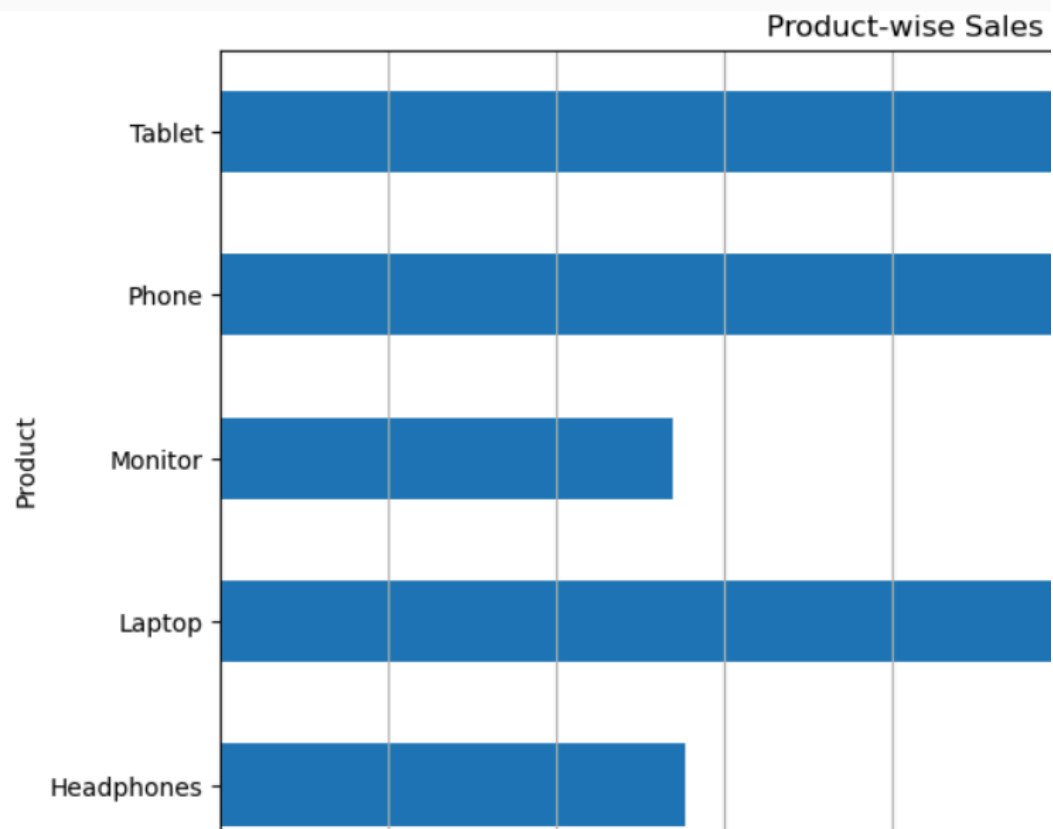
Analysis: Confirms the overall monthly trend.

Sales by Region



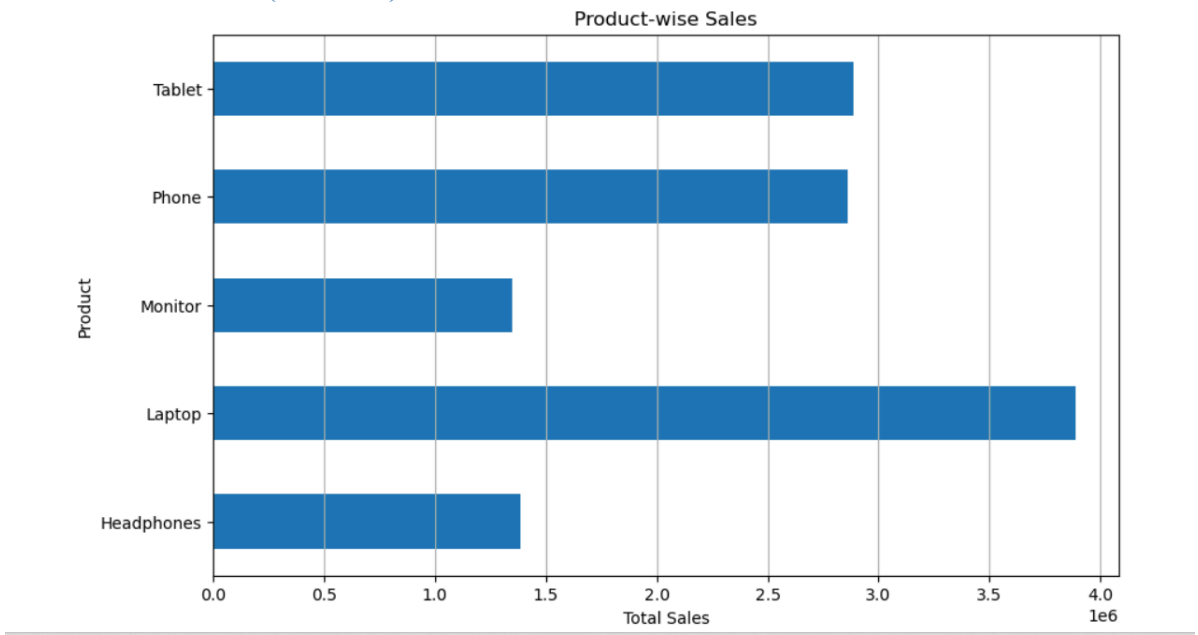
Analysis: North and South contribute the highest share of revenue.

Product-wise Sales



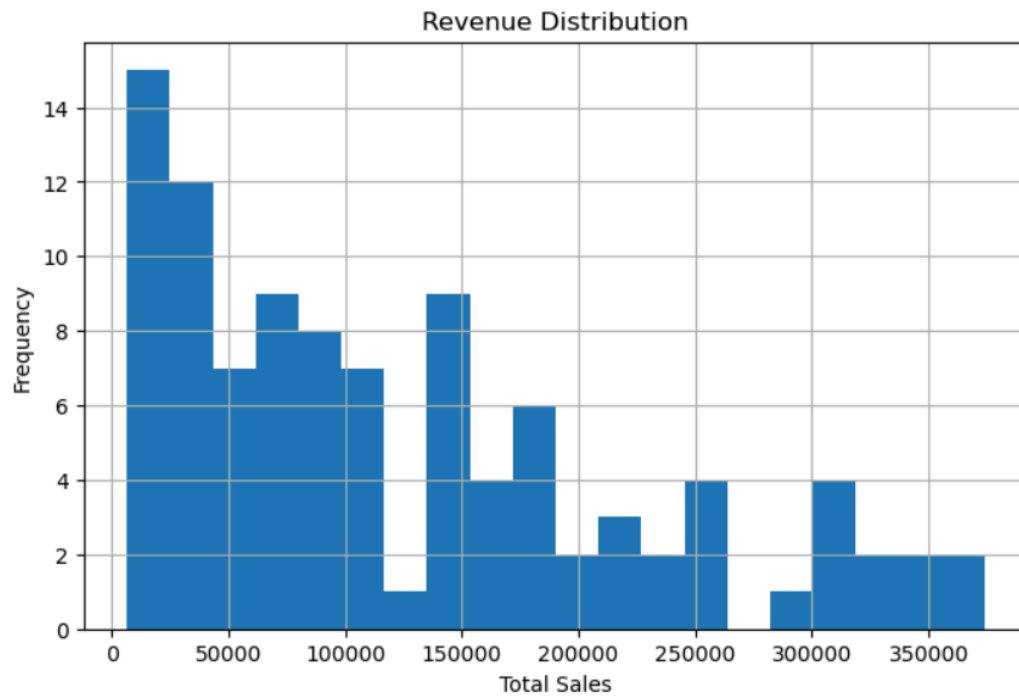
Analysis: Laptop records the highest sales among products.

Product-wise Sales (Detailed)



Analysis: Horizontal comparison of product revenue.

Revenue Distribution



Analysis: Most transactions are low-to-medium value with fewer high-value sales.

Key Findings

- March achieved the highest monthly sales.
- April recorded the lowest sales.
- Laptop is the best-selling product.
- North and South regions generated the largest share of revenue.
- A small number of customers contributed a large portion of sales.

Business Recommendations

Increase marketing in high-performing regions, retain top customers through loyalty programs, maintain inventory for laptops and phones, and improve sales in low-performing regions.

Challenges Faced

Resolved column name mismatches, datetime conversion issues, grouping errors, pivot table creation and plotting issues.

Conclusion

The project successfully demonstrates customer sales analysis using Pandas. Data cleaning, aggregation and visualization produced actionable business insights for decision-making.